



Fig. 1: Power vs frequency display with different RBW settings

ple, if the crystal stability is $1\text{E-}7/\text{day}$, with option 01 it may be $1\text{E-}9/\text{day}$. So while ordering a higher stability, specify the option number.

Sensitivity. This gives the minimum required signal for an instrument to function. It is defined for instruments that accept signal for making measurements. Frequency counters, power meters, voltmeters and spectrum analysers are such types of instruments.

In the case of a spectrum analyser, sensitivity is a measure of the ability to detect and display a small signal, and it is equal to the noise generated by the instrument itself. A signal level equal to the noise level will give a 3dB display. Signal levels less than the sensitivity cannot be detected.

Sensitivity of spectrum analysers depends on the resolution bandwidth. Sensitivity at 10kHz resolution bandwidth will be 10 dB better than sensitivity at 100kHz resolution bandwidth. So while defining sensitivity of spectrum analysers, resolution bandwidth is also to be specified.

Sensitivity of spectrum analysers is also called 'displayed average noise level.' It can be increased with the use of an additional preamplifier supplied by the manufacturer. The preamplifier will have a very low noise figure compared to the spectrum analyser, thus increasing sensitivity.

In the case of frequency counters, sensitivity is the minimum signal level at which the counter can read the frequency. For signals below this level, the counter will not respond.

For oscilloscopes, sensitivity is de-

fined in terms of deflection factor. Vertical volts/division gives sensitivity.

In the case of power meters, sensitivity is the minimum power level that can be measured. A power meter consists of a power sensor, which is connected to the power meter through a cable. The frequency range of measurement and sensitivity depend on the power sensor used. A single sensor may not cover the entire range of frequencies as well as power levels. For higher power-level measurements and very low power-level measurements, separate sensors may be required to cover the power-level range. The cost of the power sensor may be high. So select a power sensor depending upon your application requirements. Future requirements can be met by procuring an extra power sensor whenever the requirement arises.

Maximum input. This specification is needed for instruments that accept signals for making measurements. For power meters, the maximum input is the signal power level that can be fed to the power sensor without damaging the sensor. If signal level beyond maximum input is fed, the sensor may get permanently damaged. In spectrum analysers, the maximum input signal is the level that will not damage the first mixer of the spectrum analyser. Signal level beyond it will damage the first mixer permanently and sensitivity of the spectrum analyser will be poor.

Resolution. This is the ability to resolve the two values being measured as two different and distinct values. For example, in the case of spectrum

analysers, two signals separated by a given frequency should appear as two different signals. This will happen only if frequency separation is more than resolution bandwidth (RBW) of the spectrum analyser. If frequency separation is less than resolution bandwidth, the two signals will merge and appear as a single signal.

Fig. 1 shows displays with different RBW settings. In the first display, RBW is 100 kHz and the sideband signals have merged to appear as a single signal. In the second display RBW is 10 kHz, and in the third display RBW is 100 Hz. Here the sidebands are clearly separated and resolved as different frequency components.

Thus for spectrum analysers, frequency resolution is to be specified in order to resolve different frequency components.

In the case of frequency generators, resolution is the ability to set the frequencies that are distinguishable. This is the smallest increment by which the output frequency can be changed.

Frequency resolution is an important parameter for frequency synthesisers. Frequency synthesisers are basically variable-frequency generators that are very accurately and quickly settleable and highly stable. The output frequency can be varied in steps—the smallest step being the resolution.

Frequency synthesisers are used as the reference source for phase-locked oscillators. The output of phase-locked oscillators will be some integral multiple of the reference source. For example, if the multiplying factor is 50 and the phase-locked oscillator frequency is to be changed in steps of 1000 Hz, the synthesiser should have a resolution of at least 20 Hz. Thus the resolution specification depends on the requirement.

In the case of instruments that measure power levels, resolution is the ability to distinguish two power